

Saif Shikalgar

Cybersecurity Analyst & Penetration Testing Engineer

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Education

G. H. Raisoni College of Engineering and Management

Pune, India

B.Tech in Computer Science & Engineering (Cybersecurity Specialization); **CGPA: 8.10/10.0**

Sep 2023 – May 2027 (Expected)

Coursework: Network Security & Cryptography, Information Security, Computer Networks, Operating Systems, Threat Intelligence, Penetration Testing, Data Structures & Algorithms, DBMS.

Technical Skills

Security & Penetration Testing: Wireshark, Metasploit, Exploit-DB, Nmap, Burp Suite, CVE Auditing, Packet Sniffing, 802.11 WPA2 EAPOL/PMKID Handshake Interception, Hardware Hacking, Threat Modeling (STRIDE/DREAD).

Hardware & Network Defense: Dual-Gate CAN IPS/IDS, Mahalanobis Distance Anomaly Detection, Jitter Analysis, Firewall Filtering, Cryptographic Primitives (AES, RSA, SHA-256), Zero-Trust Architecture.

Languages & Scripting: Python, C, C++ (C11/C++20), Bash, PowerShell Scripting, SQL, Kotlin, Solidity.

Operating Systems & Platforms: Linux (Kali, Ubuntu, Debian), Windows Internals & Registry, Docker, STM32, ESP32, ESP8266, GCP.

Work Experience & Engineering Leadership

Team Abhyuday Racing

Pune, India

Automotive Cybersecurity Lead – Drive-by-Wire

Jan 2025 – Present

- Engineered network-level message authentication and priority arbitration over vehicle CAN 2.0B buses, hardening communication against DoS and frame injection across distributed ECUs.
- Performed penetration testing and fuzzing assessments on vehicle diagnostic interfaces, identifying potential attack vectors prior to national deployment.
- Programmed real-time deterministic **Autonomous Emergency Braking (AEB)** control logic, securing **1st Place Nationally at aBAJA 2026** with zero safety incidents.

AuraByte Studios (Independent Venture)

Pune, India

Lead Application Security Engineer & Founder

Dec 2025 – Present

- Implemented secure token-based authentication and zero-trust data persistence preventing local file tampering across **5.73K+ total installs** and **2.09K+ active devices**.
- Deployed serverless microservices on **GCP Cloud Functions** with cryptographically signed webhook validation for Google Play Billing transactions.

Flagship Cybersecurity Projects

Suraksha360: Plug-and-Play IoT Security Auditor & CVE Scanner | C++, ESP8266, Arduino Pro Micro, Exploit-DB

- Created a plug-and-play USB HID hardware security auditing tool that executes automated zero-install system vulnerability scans upon connection to Windows hosts.
- Extracted installed software registries, firewall configurations, and open ports; cross-referenced versions against Exploit-DB for known CVE vulnerabilities and streamed real-time risk scores to an ESP8266 dashboard.

Dual-Gate STM32 Hardware CAN Firewall & Anomaly IDS/IPS | STM32F446RE, C11, CMSIS/HAL, FreeRTOS, CAN 2.0B

- Engineered an inline dual-transceiver CAN bus firewall intercepting OBD-II injection, fuzzing, and RPM spoofing attacks with a deterministic **5.92 μ s processing latency** (1,065 clock cycles at 180 MHz).
- Implemented fixed-point Mahalanobis distance anomaly detection and jitter statistical analysis, rejecting **88.5% of noise** and achieving a **99%+ threat mitigation rate**.

Wireless Penetration Testing Toolkit | ESP32, C++, 802.11, Wireshark, PCAP

- Engineered a portable 802.11 wireless auditing tool on ESP32 capable of passive beacon sniffing, targeted deauthentication, and WPA2 EAPOL/PMKID four-way handshake capture.
- Logged captured wireless packets directly to PCAP files on microSD storage for deep-packet inspection in Wireshark; implemented rogue AP detection.

CYO: Privacy-First Multimodal AI Search Engine | PyTorch, CLIP, FAISS, Docker, CUDA

- Built an air-gapped local semantic image search platform keeping all image vectorizations and queries 100% on-device with zero external network leakage.

Honors & Awards

1st Place, Autonomous Emergency Braking (AEB): National aBAJA 2026 Competition; verified safety-critical embedded architecture.

1st Place, Manufacturing Excellence Award: National aBAJA 2026 Competition; recognized for robust fail-safe system design.

1st Place, Adaptive Cruise Control (ACC): aBAJA 2025; achieved following an 8-hour high-stakes crisis hardware migration.